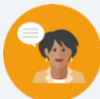


3.7 Solving Problems Using Angle Pair Relationships

Lesson Introduction

 Benchmark	MA.912.GR.1.1: Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of lines and angles.		 Learning Target	I can solve mathematical and real-world problems involving angle pair relationships formed by transversals and parallel lines.
 Benchmark Coherence	Building On MA.8.GR.1.4 Solve mathematical problems involving the relationships between supplementary, complementary, vertical, or adjacent angles.		Working Toward N/A	
 Essential Questions	How can you use the angle relationships, postulates, and theorems to solve problems?			
 Lesson Narrative	Students collaborate to solve for variables and missing angles using angle relationships, postulates, and theorems given either mathematical or real-world problems. This lesson is split between partner work and individual work as students apply their knowledge of all angle relationships learned in the unit to solve problems.			
 Component Summary	3.7.1 Warm-Up (~5 min.)	Match angle pairs based on their relationship in a map (<i>SE p. 158</i>)	3.7.3 Your Turn! (15-20 min.)	Practice using angle pair relationships to solve problems (<i>SE p. 161</i>)
	3.7.2 Collaboration (15-20 min.)	Use angle relationships to solve for variables and unknown angles (<i>SE p. 159</i>)	3.7.4 Wrap-Up (~5 min.)	Reflect on learning targets using a scale (<i>SE p. 163</i>)
 Resources	Glossary		Materials	
	Key Terms: N/A Additional Terms: - Congruent - Supplementary Angles - Parallel Lines - Transversal		(Optional) Highlighters or Colored Pencils	
			Additional Preparation N/A	

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may use the incorrect angle pair relationships when creating equations to solve. - Students may make errors related to solving equations. - Students may confuse when they are solving for a variable versus a missing angle.	Consider how you will use the data collected from the 3.6.4 Wrap-Up as well as observational data from Lesson 6 to determine the supports to give to students during this lesson.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
 Differentiate Instruction	Think-Pair-Share Consider using this routine in the 3.7.2 Collaboration. The activity is set up to be completed with a partner, but by adding the Think-Pair-Share routine and allowing students to have alone time to work prior to working with their partner, richer discussion that benefit the pair and the class as a whole are more likely to occur.	MA.K12.MTR.3.1: Mathematical Fluency The components of this lesson focus on student practice applying what they have learned thus far in Unit 3. Students should select efficient and appropriate methods for solving problems within the given mathematical or real-world context. Provide students with the flexibility to solve problems by selecting a procedure that allows them to solve efficiently and accurately.
	3.7.2 Collaboration provides both auditory and visual entry points into discussion. Consider providing roles for students that incorporate multiple entry points by making one student the writer as the other student tells what to write to solve the problem. Students should swap roles for each problem to allow both students the opportunity to participate in each role.	
Lesson Notes:		

3.7.1 Warm-Up: Angle Pair Relationships

Component Overview: Students will use a given map to complete this component by naming angle pairs. Students will match different points of interest on the map with the angle relationship the locations form.



3.7.1 Warm-Up: Angle Pair Relationships

- The image shows two streets intersected by a third street. Various points of interest are located at the intersections.



Match the listed points of interest with their angle relationship listed in the right column. Write the letter of the angle relationship on the line.

Points of Interest	Angle Relationship
<u>E</u> Museum and Worship Center	A. Alternate Interior Angles
<u>A</u> Bank and Day Care	B. Consecutive Interior Angles
<u>D</u> Park and Coffee Shop	C. Corresponding Angles
<u>F</u> Fire Station and Worship Center	D. Vertical Angles
<u>B</u> Community Center and Day Care	E. Alternate Exterior Angles
<u>C</u> Bank and Museum	F. Linear Pair



If time permits, ask students to confer with a partner about which angle pairs they struggle with identifying. Provide partners the opportunity to share strategies with one another and answer their peers' questions.

3.7.2 Collaboration: Solving Problems Using Angle Relationships

Component Overview: Students work with a partner to solve mathematical and real-world problems by applying angle relationships, postulates, and theorems in context.



3.7.2 Collaboration: Solving Problems Using Angle Relationships

Work with your partner to solve each problem.

1. Mako is a steel roller coaster located at Sea World in Orlando. Some of the support beams are shown in the diagram using $\overline{KL}$, $\overline{LR}$, and $\overline{FR}$.



Determine the value of x that shows that $\overline{KL} \parallel \overline{FR}$, if $m\angle KLR = (7x - 5.6)^\circ$ and $m\angle FRL = (3x + 5.6)^\circ$.
 $x = 2.8$



When solving multiple-step problems involving angle relationships, consider having students highlight patterns they have seen in the procedures and use them to build their own scaffolding support.

3.7.2 Collaboration: Solving Problems Using Angle Relationships (continued)

2. Most of the streets in Miami are on a grid. Beacom Blvd. is one of the few streets that does not run north to south or east to west. The diagram shows a map of Beacom Blvd., $\overline{HG}$, between SW 4th Street, $\overline{WT}$, and SW 6th Street, $\overline{BR}$, just a few blocks from Miami Senior High School.



- a. Given that $\overline{WT} \parallel \overline{BR}$, determine the value of x if $m\angle HMB = (2x - 7)^\circ$, $m\angle WKG = (4x + 25)^\circ$, and $m\angle KMB = (6x - 29)^\circ$.
 $x = 27$

- b. Determine the measure of each angle.

$$m\angle HMB$$

$$m\angle HMB = 47^\circ$$

$$m\angle HMR$$

$$m\angle HMR = 133^\circ$$

$$m\angle TKM$$

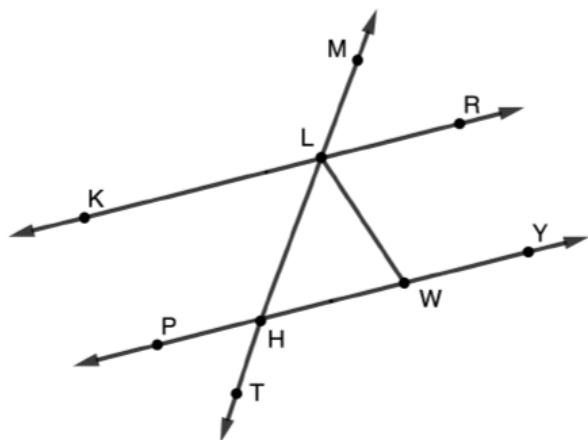
$$m\angle TKM = 133^\circ$$



For question 2a, there is more than one way to solve for x based on the given information. While students are working, observe and make note of which students used different methods. Then, after students have completed the question, select students to share their method. Encourage students to provide justification for the steps in their method to provide clarity for those students who are still struggling seeing the angle relationships.

3.7.2 Collaboration: Solving Problems Using Angle Relationships (continued)

3. $\overleftrightarrow{KR}$, $\overleftrightarrow{PY}$, and $\overleftrightarrow{MT}$ are shown with $\overleftrightarrow{LW}$ where $m\angle LWH = (4y - 9\frac{1}{3})^\circ$, $m\angle WLH = (3y + 3)^\circ$, and $m\angle LHW = m\angle WLH$.



- a. Determine the value of y that shows that $\overleftrightarrow{PY} \parallel \overleftrightarrow{KR}$.
 $y = \frac{55}{3}$
- b. Determine the measure of each angle.

$$m\angle RLW$$

$$m\angle RLW = 64^\circ$$

$$m\angle PHT$$

$$m\angle PHT = 58^\circ$$

$$m\angle KLM$$

$$m\angle KLM = 122^\circ$$



Students may not recognize that $\overleftrightarrow{LW}$ is also a transversal of $\overleftrightarrow{KR}$ and $\overleftrightarrow{PY}$. If they miss this relationship, they may struggle with this problem. Listen and watch as students work with their partner to determine whether to provide additional scaffolding support.

For students who struggle to identify the angle pair relationships, provide the students with two tables for them to draw the parallel lines twice, one with transversal $\overleftrightarrow{MT}$ and one with transversal $\overleftrightarrow{LW}$. Then, in the table with transversal $\overleftrightarrow{MT}$, have the students name all the angles that are congruent to $\angle MLR$ and those congruent to $\angle MLK$. Then, in the table with transversal $\overleftrightarrow{LW}$, have them name those angles congruent to $\angle RLW$ and those congruent to $\angle KLW$. Providing this visual entry-point may help some students to determine which angle relationship to use with each angle pair.

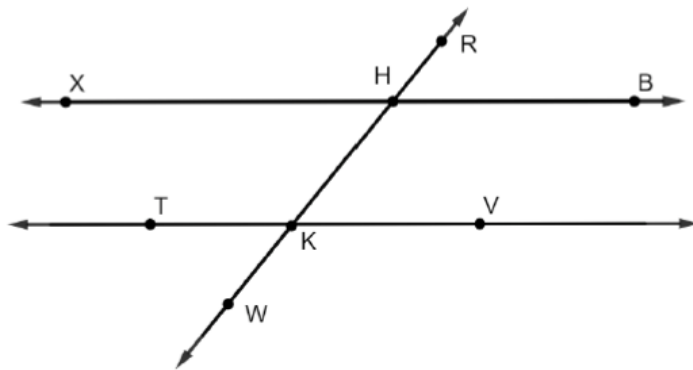
3.7.3 Your Turn!

Component Overview: Students independently apply angle relationships, postulates, and theorems to solve mathematical and real-world problems.



3.7.3 Your Turn!

1. $\overleftrightarrow{XB}$, $\overleftrightarrow{TV}$, and $\overleftrightarrow{RW}$ are shown. $\overleftrightarrow{XB} \parallel \overleftrightarrow{TV}$ where $\overleftrightarrow{RW}$ is a transversal with $m\angle XHR = (9x + 21)^\circ$ and $m\angle HKV = (4x + 3)^\circ$.



- a. Determine the value of x .

$$x = 12$$

- b. Find the measure of each angle.

$$m\angle XHR \\ 129^\circ$$

$$m\angle HKV \\ 51^\circ$$

$$m\angle TKW \\ 51^\circ$$

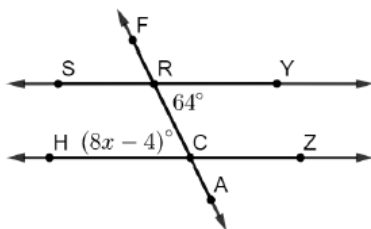
$$m\angle BHK \\ 129^\circ$$



For students who struggle setting up the equations, have them first highlight or name all the congruent angles to $\angle RHB$ and those congruent to $\angle RHX$. Then, have them state the relationship of angle pairs in the image that are not congruent to each other.

3.7.3 Your Turn! (continued)

2. $\overleftrightarrow{SY} \parallel \overleftrightarrow{HZ}$, $m\angle YRC = 64^\circ$, and $m\angle RCH = (8x - 4)^\circ$.



- a. Find the value of x .

$$x = 8.5$$

- b. Find the measure of each angle.

$$m\angle ZCA$$

$$64^\circ$$

$$m\angle FRS$$

$$64^\circ$$

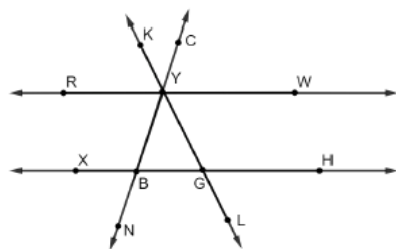
$$m\angle YRF$$

$$116^\circ$$

$$m\angle HCA$$

$$116^\circ$$

3. $\overleftrightarrow{RW} \parallel \overleftrightarrow{XH}$, $m\angle RYK = s^\circ$, $m\angle CYW = m^\circ$, $m\angle GYB = 43^\circ$, $m\angle YBG = 72^\circ$, and $m\angle LGH = (6t + 11)^\circ$.



Find the value of each variable. Show your work.

t

$$6t + 11 + 72 + 43 = 180$$

$$t = 9$$

m

$$m = 72^\circ$$

s

$$s = (6t + 11)^\circ$$

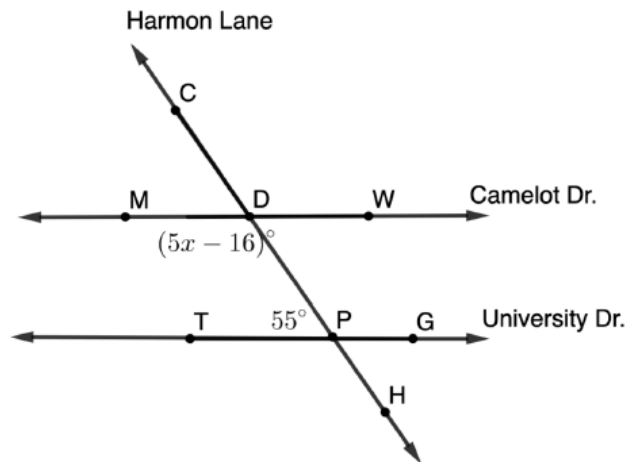
$$s = 65^\circ$$



If students struggled with question 1, consider working with them individually or in small groups on question 2 to clear up misconceptions and to provide remediated support.

3.7.3 Your Turn! (continued)

4. Harmon Lane, $\overleftrightarrow{CH}$, crosses the streets Camelot Drive, $\overleftrightarrow{MW}$, and University Drive, $\overleftrightarrow{TG}$, as shown. $m\angle TPD = 55^\circ$, and $m\angle MDP = (5x - 16)^\circ$.



Determine the value of x that shows $\overleftrightarrow{MW} \parallel \overleftrightarrow{TG}$.

$x = 28.2$



Challenge students who finish early to write at least two possible assessment questions, one mathematical and one real-world, that involve angle pair relationships formed by two parallel lines and a transversal. Encourage students to be creative, use values and expressions, and provide an answer key. Consider using the questions on a future assessment and award points or credit to any students whose questions are used.

3.7.4. Wrap-Up: Reflection

Component Overview: Students use a scale to reflect on learning targets related to solving problems involving different angle relationships.



3.7.4 Wrap-Up: Reflection

Draw an **X** on each line to indicate where your current learning is related to each target.

Answers will vary.

1. I can use corresponding angles to determine angle measurements.

I need help.
I can't get started.

I'm getting there, but
I need more practice.

I understand this,
and I feel confident.



2. I can use alternate interior angles to determine angle measurements.

I need help.
I can't get started.

I'm getting there, but
I need more practice.

I understand this,
and I feel confident.



3. I can use alternate exterior angles to determine angle measurements.

I need help.
I can't get started.

I'm getting there, but
I need more practice.

I understand this,
and I feel confident.



4. I can use consecutive interior angles to determine angle measurements.

I need help.
I can't get started.

I'm getting there, but
I need more practice.

I understand this,
and I feel confident.



5. I can use consecutive exterior angles to determine angle measurements.

I need help.
I can't get started.

I'm getting there, but
I need more practice.

I understand this,
and I feel confident.



The information provided from the student here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons.